

**SECTION 1. PRODUCT IDENTIFICATION**

Product name : LIQUID LAUNDRY SOUR

Other means of identification : Not applicable

Recommended use : Laundry product

Product dilution information : 0,02 % - 0,1 %

Company : Distributed in the Caribbean by:  
Carr. #28 Lote 13-A Urbanización Industrial Luchetti Bayamón, PR  
00961

Emergency telephone : Caribbean:787-796-1290 x-6855 / 787-599-6743/1-800-984-1290

Issuing date : 26.05.2021

**SECTION 2. HAZARD OR HAZARDS IDENTIFICATION**
**GHS Classification**
**Product AS SOLD**

Skin corrosion/irritation, Sub-category 1A

H314

Serious eye damage/eye irritation, Category 1

H318

**Product AT USE DILUTION**

Not a hazardous substance or mixture.

**GHS label elements**
**Product AS SOLD**

Hazard pictograms



Signal Word : Danger

Hazard Statements : H314 Causes severe skin burns and eye damage.

Precautionary Statements : **Prevention:**  
P264 Wash skin thoroughly after handling.  
P280 Wear protective gloves/ protective clothing/ eye protection/ face protection.

**Response:**

P301 + P330 + P331 IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.

P303 + P361 + P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water.

P304 + P340 + P310 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Immediately call a POISON CENTER/ doctor.

P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor.

P363 Wash contaminated clothing before reuse.

## SAFETY DATA SHEET

### LIQUID LAUNDRY SOUR

**Storage:**

P405

Store locked up.

**Disposal:**

P501

Dispose of contents/ container to an approved waste disposal plant.

**Product AT USE DILUTION**

Precautionary Statements

**Prevention:**

P264

Wash hands thoroughly after handling.

**Response:**

P314

Get medical advice/ attention if you feel unwell.

**Storage:**

P401

Store in accordance with local regulations.

**Product AS SOLD****Other hazards**

: Do not mix with bleach or other chlorinated products – will cause chlorine gas.

### SECTION 3. COMPOSITION / INFORMATION OF HAZARDOUS INGREDIENTS

**Product AS SOLD**

Pure substance/mixture

: Mixture

**Chemical name**

formic acid

**CAS-No.**

64-18-6

**Concentration (%)**

10 - 30

citric acid

77-92-9

1 - 5

**Product AT USE DILUTION**

No hazardous ingredients

### SECTION 4. FIRST AID MEASURES

**Product AS SOLD**

In case of eye contact

: Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact

: Wash off immediately with plenty of water for at least 15 minutes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

If swallowed

: Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

If inhaled

: Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

Protection of first-aiders

: If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician

: Treat symptomatically.

Most important symptoms and effects, both acute and delayed

: See Section 11 for more detailed information on health effects and symptoms.

## SAFETY DATA SHEET

### LIQUID LAUNDRY SOUR

#### Product AT USE DILUTION

|                         |                                                         |
|-------------------------|---------------------------------------------------------|
| In case of eye contact  | : Rinse with plenty of water.                           |
| In case of skin contact | : Rinse with plenty of water.                           |
| If swallowed            | : Rinse mouth. Get medical attention if symptoms occur. |
| If inhaled              | : Get medical attention if symptoms occur.              |

#### Potential Health Effects

##### Product AS SOLD

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Causes serious eye damage.                                  |
| Skin             | : Causes severe skin burns.                                   |
| Ingestion        | : Causes digestive tract burns.                               |
| Inhalation       | : May cause nose, throat, and lung irritation.                |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

#### Product AT USE DILUTION

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Health injuries are not known or expected under normal use. |
| Skin             | : Health injuries are not known or expected under normal use. |
| Ingestion        | : Health injuries are not known or expected under normal use. |
| Inhalation       | : Health injuries are not known or expected under normal use. |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

### SECTION 5. FIRE-FIGHTING MEASURES

#### Product AS SOLD

|                                                |                                                                                                                                                                                 |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Suitable extinguishing media                   | : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                                                                       |
| Unsuitable extinguishing media                 | : None known.                                                                                                                                                                   |
| Specific hazards during fire fighting          | : Not flammable or combustible.                                                                                                                                                 |
| Hazardous combustion products                  | : Decomposition products may include the following materials:<br>Carbon oxides<br>Nitrogen oxides (NOx)<br>Sulfur oxides<br>Oxides of phosphorus                                |
| Special protective equipment for fire-fighters | : Use personal protective equipment.                                                                                                                                            |
| Specific extinguishing methods                 | : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes. |

## SAFETY DATA SHEET

### LIQUID LAUNDRY SOUR

#### SECTION 6. MEASURES TO BE TAKEN IN CASE OF ACCIDENTAL SPILLAGE

##### Product AS SOLD

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

##### Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed.

Environmental precautions : No special environmental precautions required.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

#### SECTION 7. HANDLING AND STORAGE

##### Product AS SOLD

Advice on safe handling : Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe dust/ fume/ gas/ mist/ vapors/ spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not mix with bleach or other chlorinated products – will cause chlorine gas. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep away from strong bases. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Storage temperature : 0 °C to 50 °C

##### Product AT USE DILUTION

Advice on safe handling : Wash hands after handling. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE). For personal protection see section 8.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

#### SECTION 8. EXPOSURE CONTROLS / PERSONAL PROTECTION

##### Product AS SOLD

##### Ingredients with workplace control parameters

## SAFETY DATA SHEET

### LIQUID LAUNDRY SOUR

| Components  | CAS-No. | Form of exposure | Permissible concentration      | Basis     |
|-------------|---------|------------------|--------------------------------|-----------|
| formic acid | 64-18-6 | TWA              | 5 ppm                          | CR OEL    |
|             |         | STEL             | 10 ppm                         | CR OEL    |
| formic acid | 64-18-6 | TLV-TWA          | 5 ppm                          | DO OEL    |
| formic acid | 64-18-6 | CCT              | 10 ppm<br>19 mg/m <sup>3</sup> | PA OEL    |
|             |         | CPT              | 5 ppm<br>9 mg/m <sup>3</sup>   | PA OEL    |
| formic acid | 64-18-6 | TWA              | 5 ppm                          | ACGIH     |
|             |         | STEL             | 10 ppm                         | ACGIH     |
|             |         | TWA              | 5 ppm<br>9 mg/m <sup>3</sup>   | NIOSH REL |
|             |         | TWA              | 5 ppm<br>9 mg/m <sup>3</sup>   | OSHA Z1   |

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

#### Personal protective equipment

Eye protection : Safety goggles  
Face-shield

Hand protection : Wear the following personal protective equipment:  
Standard glove type.  
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

#### Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

#### Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

### SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

## SAFETY DATA SHEET

### LIQUID LAUNDRY SOUR

|                                         | Product AS SOLD                                | Product AT USE DILUTION |
|-----------------------------------------|------------------------------------------------|-------------------------|
| Physical State                          | : liquid                                       | liquid                  |
| Color                                   | : clear, light blue                            | colorless               |
| Odor                                    | : pungent                                      | odorless                |
| pH                                      | : 0,7 - 1,0, (100 %)                           | 5,5 - 7,5               |
| Flash point                             | : Not applicable, Does not sustain combustion. |                         |
| Odor Threshold                          | : No data available                            |                         |
| Melting point/freezing point            | : No data available                            |                         |
| Initial boiling point and boiling range | : No data available                            |                         |
| Evaporation rate                        | : No data available                            |                         |
| Flammability (solid, gas)               | : Not applicable                               |                         |
| Upper explosion limit                   | : No data available                            |                         |
| Lower explosion limit                   | : No data available                            |                         |
| Vapor pressure                          | : No data available                            |                         |
| Relative vapor density                  | : No data available                            |                         |
| Relative density                        | : 1,036 - 1,056                                |                         |
| Water solubility                        | : soluble                                      |                         |
| Solubility in other solvents            | : No data available                            |                         |
| Partition coefficient: n-octanol/water  | : No data available                            |                         |
| Autoignition temperature                | : No data available                            |                         |
| Thermal decomposition                   | : No data available                            |                         |
| Viscosity, kinematic                    | : No data available                            |                         |
| Explosive properties                    | : No data available                            |                         |
| Oxidizing properties                    | : No data available                            |                         |
| Molecular weight                        | : No data available                            |                         |
| VOC                                     | : No data available                            |                         |

### SECTION 10. STABILITY AND REACTIVITY

#### Product AS SOLD

|                                    |                                                                                   |
|------------------------------------|-----------------------------------------------------------------------------------|
| Reactivity                         | : No dangerous reaction known under conditions of normal use.                     |
| Chemical stability                 | : Stable under normal conditions.                                                 |
| Possibility of hazardous reactions | : Do not mix with bleach or other chlorinated products – will cause chlorine gas. |
| Conditions to avoid                | : None known.                                                                     |
| Incompatible materials             | : Bases<br>Metals                                                                 |
| Hazardous decomposition products   | : In case of fire hazardous decomposition products may be produced such as:       |

## SAFETY DATA SHEET

### LIQUID LAUNDRY SOUR

Carbon oxides  
Nitrogen oxides (NOx)  
Sulfur oxides  
Oxides of phosphorus

#### SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

##### Experience with human exposure

###### Product AS SOLD

Eye contact : Redness, Pain, Corrosion  
Skin contact : Redness, Pain, Corrosion  
Ingestion : Corrosion, Abdominal pain  
Inhalation : Respiratory irritation, Cough

###### Product AT USE DILUTION

Eye contact : No symptoms known or expected.  
Skin contact : No symptoms known or expected.  
Ingestion : No symptoms known or expected.  
Inhalation : No symptoms known or expected.

##### Toxicity

###### Product AS SOLD Product

Acute oral toxicity : No data available  
Acute inhalation toxicity : No data available  
Acute dermal toxicity : No data available  
Skin corrosion/irritation : No data available  
Serious eye damage/eye irritation : No data available  
Respiratory or skin sensitization : No data available  
Carcinogenicity : No data available  
Reproductive effects : No data available  
Germ cell mutagenicity : No data available  
Teratogenicity : No data available  
STOT-single exposure : No data available  
STOT-repeated exposure : No data available  
Aspiration toxicity : No data available

###### Components

Acute oral toxicity : citric acid  
LD50 Rat: 11.700 mg/kg

## SAFETY DATA SHEET

### LIQUID LAUNDRY SOUR

#### SECTION 12. ECOLOGICAL INFORMATION

##### Ecotoxicity

###### Product AS SOLD

Environmental Effects : This product has no known ecotoxicological effects.

###### Product AT USE DILUTION

Environmental Effects : This product has no known ecotoxicological effects.

###### Product AS SOLD

###### Product

Toxicity to fish : No data available

Toxicity to daphnia and other aquatic invertebrates : No data available

Toxicity to algae : No data available

##### Components

Toxicity to fish : formic acid  
96 h LC50: > 100 mg/l  
  
citric acid  
96 h LC50 Fish: > 100 mg/l

##### Persistence and degradability

###### Product AT USE DILUTION

Readily biodegradable.

##### Bioaccumulative potential

No data available

##### Mobility in soil

No data available

##### Other adverse effects

No data available

#### SECTION 13. INFORMATION REGARDING THE DISPOSAL OF PRODUCTS

###### Product AS SOLD

Disposal methods : Avoid that the product entering in the body of water, drains, pipes, or the earth (grounds). Where possible, recycling is preferable to disposal or incineration in accordance with local legislation. If you can't recycle, dispose according to local regulations. Elimination of waste in approved waste disposal plants.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations. Remove as unused products. Empty containers should be brought to local recycling or waste disposal according to local legislation. Do not reuse empty containers.

###### Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.



## SAFETY DATA SHEET

### LIQUID LAUNDRY SOUR

#### SECTION 14. TRANSPORT INFORMATION

##### Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

##### Land transport

UN number : 3412  
Description of the goods : FORMIC ACID  
Class : 8  
Packing group : II  
Environmentally hazardous : no

##### Air transport (IATA)

Contact Regulatory for air freight eligibility

##### Sea transport (IMDG/IMO)

UN number : 3412  
Description of the goods : FORMIC ACID  
Class : 8  
Packing group : II  
Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code : Not applicable  
Marine pollutant : no

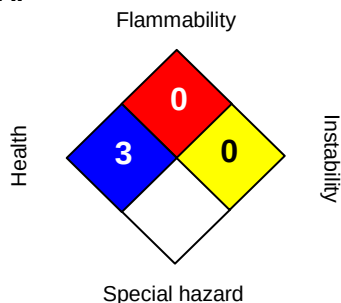
#### SECTION 15. REGULATORY INFORMATION

Our Safety Data Sheet (SDS) complies with Decree n ° 40705-S

#### SECTION 16. OTHER INFORMATION

##### Product AS SOLD

###### NFPA:



###### HMIS III:

|                 |   |
|-----------------|---|
| HEALTH          | 3 |
| FLAMMABILITY    | 0 |
| PHYSICAL HAZARD | 0 |

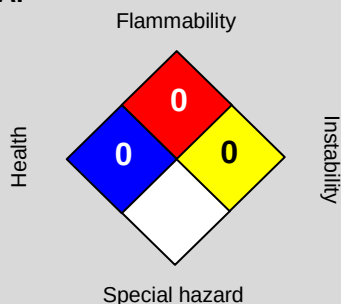
0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

##### Product AT USE DILUTION

## SAFETY DATA SHEET

### LIQUID LAUNDRY SOUR

#### NFPA:



#### HMIS III:

|                        |          |
|------------------------|----------|
| <b>HEALTH</b>          | <b>0</b> |
| <b>FLAMMABILITY</b>    | <b>0</b> |
| <b>PHYSICAL HAZARD</b> | <b>0</b> |

0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

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Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.